BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 284 · Packet switching, circuit switching and standards

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which TCP/IP layer contains TCP?

Q2. What does DNS usually provide when given a domain name?

Q3. Can a graph have more than one possible path between two vertices?

Q4. Name one advantage of sharing a network link instead of reserving it for one user.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

packet switching — Data is split and forwarded across shared network links.

circuit switching — Communication uses a pre-established reserved path.

sequence information — Control information that allows ordered reconstruction.

retransmission — Sending missing or unacknowledged data again.

standard — Agreed specification supporting compatibility and interoperability.

New learning

Split a message into packets — The sender can split data into separate packets or segments. Each includes control information such as addressing and/or sequence information at relevant protocol layers.

Packets may take different routes — Packet switching lets network traffic share links. IP forwarding can choose different paths; it does not guarantee the same arrival order.

Restore order when needed — TCP provides a reliable ordered byte stream, using acknowledgements and retransmission if required. Do not confuse TCP guarantees with best-effort IP delivery.

Worked example

Two TCP segments with sequence positions 1 and 2 travel on different routes and arrive 2, then 1. Explain the final data order.

1. Route — The network may forward them along different paths, so arrival order is not guaranteed.

2. Receive — TCP uses sequence information to buffer out-of-order data.

3. Deliver — When missing earlier bytes arrive, TCP makes data available to the application in the correct order: first 1, then 2.

We Do · Practise with guidance

Three TCP data segments arrive as 1, 3, with segment 2 missing. Explain what TCP can do.

Detect gap

The receiver recognises that data between segment 1 and segment 3 is missing.

Recover

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

TCP can cause the missing data to be retransmitted; once the gap is filled, data is delivered in order.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Give one difference between packet switching and circuit switching. [2]

Question 2

Data labelled 1,2,3 arrives as 2,1,3 using TCP. Explain how TCP handles the out-of-order arrival and what the application receives. [4]

Question 3

Explain one advantage of packet switching and why communication standards are important for different manufacturers' devices. [4]

Exit Ticket · Answer before checking

IP guarantees that packets always arrive in the order sent.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Transport layer.

Q2. An IP address for the name, when the appropriate DNS records exist.

Q3. Yes.

Q4. Capacity can be used by other traffic when a user is idle, improving efficiency.

You Do pages · Independent questions

Q1. [2] Packet switching forwards separate packets using shared network links [1]; circuit switching establishes a dedicated or reserved connection/path resources for a session [1].

Q2. [4] TCP identifies the out-of-order data from sequence information [1], buffers later data while earlier bytes are missing [1], releases the ordered stream once the gap is filled [1], so the application receives 1,2,3 [1]. This uses simplified labelled segments; TCP sequence numbers actually identify byte positions.

Q3. [4] Packet switching allows links to be shared efficiently or can route around a failed path [2 with explanation]; agreed communication standards define consistent protocols/formats [1], allowing independently produced devices and software to interoperate [1].

Exit Ticket · Model answer

False. IP is best effort; protocols such as TCP offer reliable ordered delivery.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding